AMERICAN UNIVERSITY OF BEIRUT
MAROUN SEMAAN FACULTY OF ENGINEERING & ARCHITECTURE

Department of Industrial Engineering and Management (IEM)

How Time And Interest Affect Money

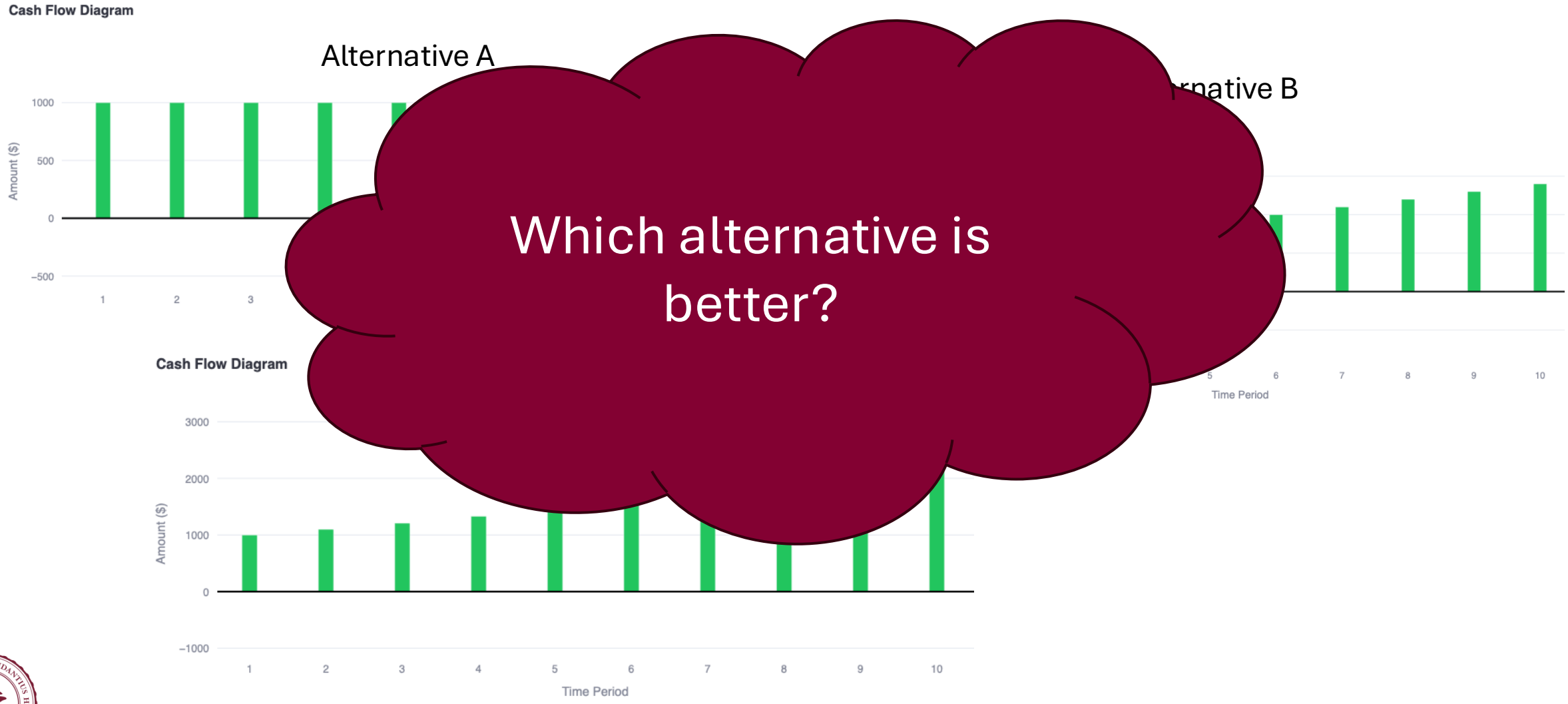
Engineering Economy

Chapter 2

Understanding Engineering Economy **Factors**



Why do we need Factors?



Why do we need Factors?



Engineering Economy Factors

Tools to manipulate the various cash flows

- Transform cash flows while considering the **time value of money**
  - **Find the Present worth of all alternatives**
  - **Transform the increasing into uniform and compare**

Notation

$(X/Y, i, n)$

- Find X amount, given:
  - Y amount
  - $i\%$ interest
  - n number of periods

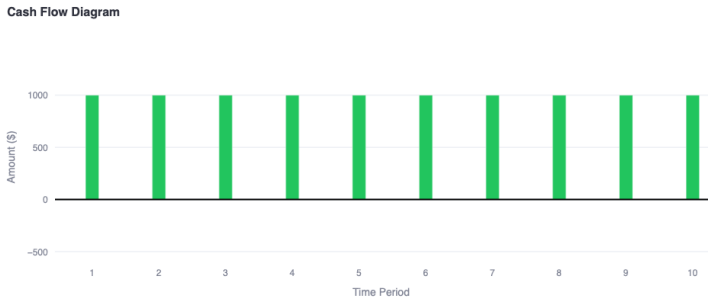
- P: Present Worth
- F: Future Worth
- A: Annual (periodic) Amount (Uniform Series)
- G: Arithmetic Gradient
- g: Geometric Gradient

Factors

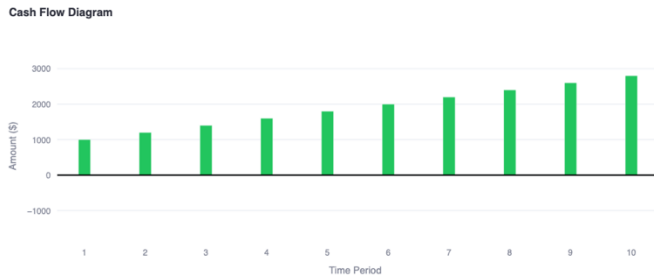
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  - Single-Payment Compound Amount (F/P)
  - Single-Payment Present Worth (P/F)



- Uniform Series Factors
  - Uniform Series Present Worth (P/A)
  - Uniform Series Compound Amount (F/A)
  - Capital Recovery (A/P)
  - Sinking Fund (A/F)



- Gradient Series
  - Arithmetic Gradient P/G, F/G, and A/G
  - Geometric Gradient

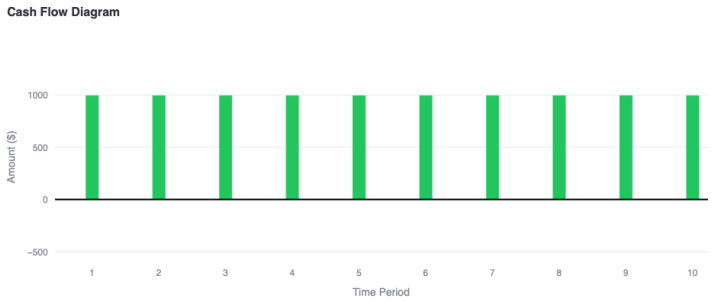


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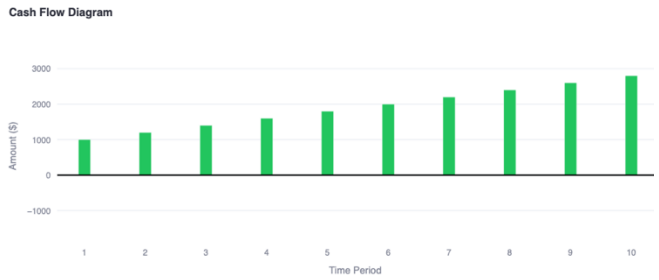
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Single Amount Factor - Single-Payment Compound Amount

$$(F/P, i, n) = (1 + i)^n$$

- Find the Future Worth given the Present Worth, the interest rate and the number of periods
- If you deposit $\$1,000$ today in a savings account that pays 5% interest per year, how much will you have in 4 years?

$$F = P(F/P, i, n) = P(F/1000, 5\%, 4)$$
$$= 1000(1 + 0.05)^4 = 1215.51$$

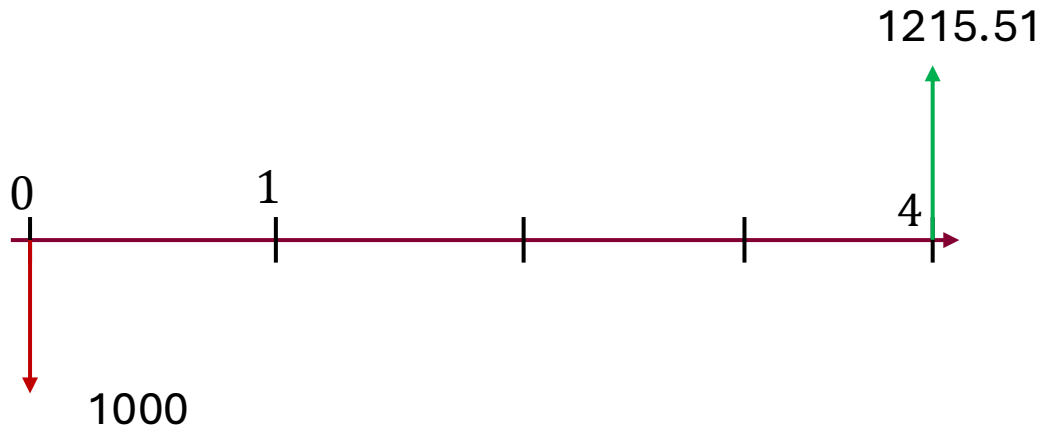


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Single Amount Factor - Single-Payment Present Worth

$$(\mathrm{P/F}, i, n) = (1+i)^{-n}$$

- Find the Present Worth given the Future Worth, the interest rate and the number of periods

	A	B	C	D	E	F
1	Present Worth	$ 10,000.00		Future Worth	Equation	$ 13,382.26
2	interest Rate	6%			Excel Fun.	$13,382.26
3	Periods	5				
4						
5						
6						
7						
8				Present Worth	Equation	$ 10,000.00
9					Excel Fun.	$10,000.00

Single Amount Factor - Excel

	A	B	C	D	E	F
1	Present Worth	10000		Future Worth	Equation	=B1*(1+B2)^B3
2	interest Rate	0.06			Excel Fun.	=FV(B2,B3,,-B1)
3	Periods	5				
4						
5						
6						
7						
8				Present Worth	Equation	=F1*(1+B2)^(-B3)
9					Excel Fun.	=PV(B2,B3,,-F2)

- FV and PV functions can be used in Excel

  PV(rate, nper, pmt, [fv], [type])

- 3rd parameters (pmt) is 0
- 4th optional parameter is negated!! Why?



Factors

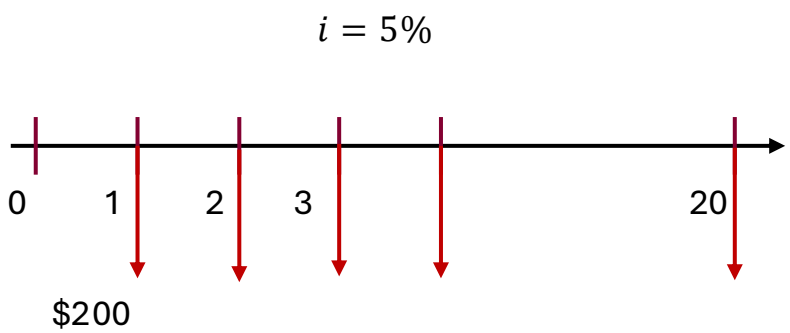
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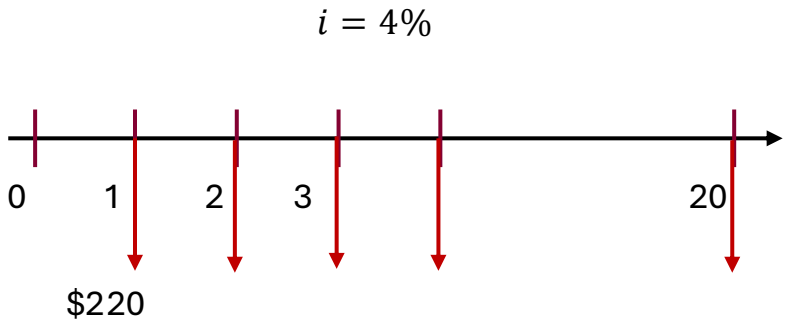
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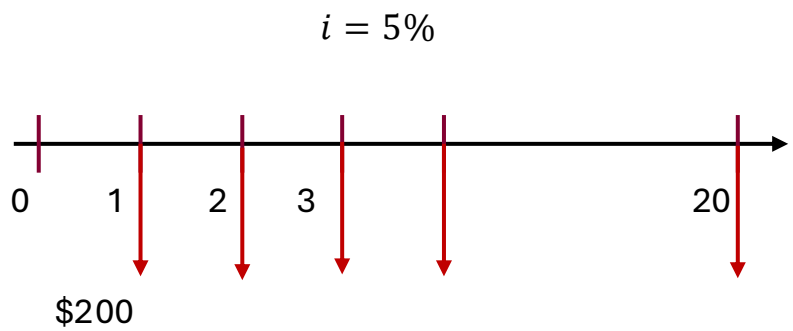
Uniform Series Factors

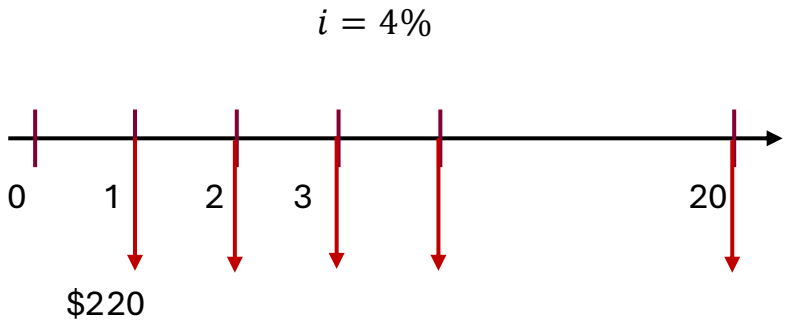




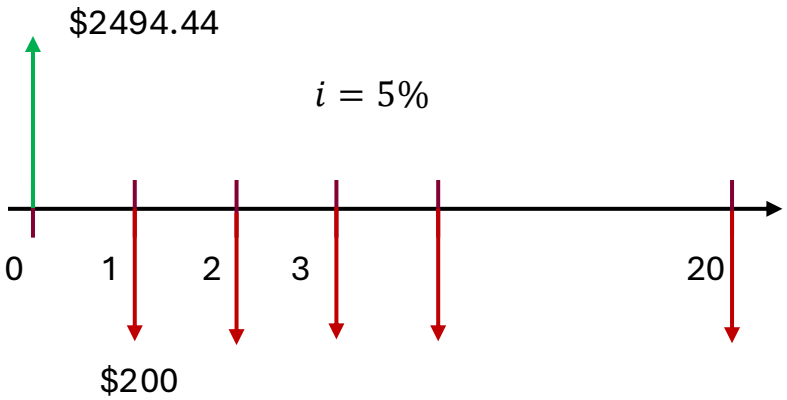
- How to compare these two alternatives?
  - Compare the Present Worth
  - Compare the Future Worth

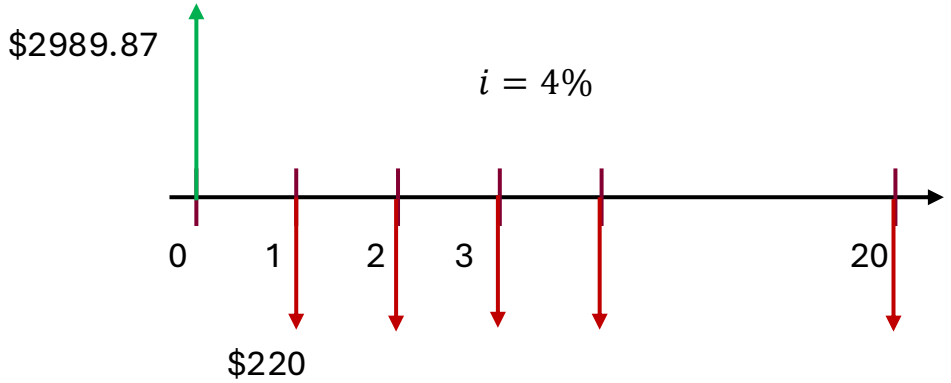
Uniform Series Factors – Present Worth Factor



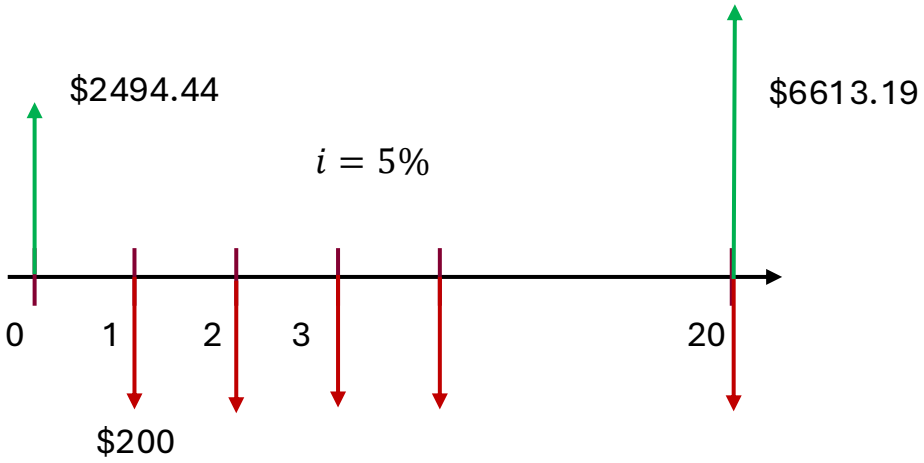


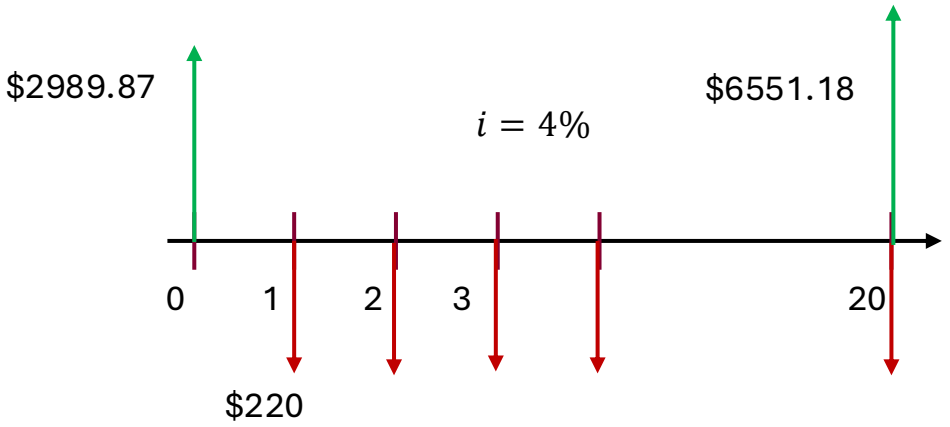
$$(P/A, i, n) = \left(\frac{(1+i)^n - 1}{i(1+i)^n}\right)$$





Uniform Series Factors – Compound Amount Factor (Future Worth)





$$(F/A) = (F/P)(P/A)$$

$$= \frac{(1+i)^n - 1}{i}$$

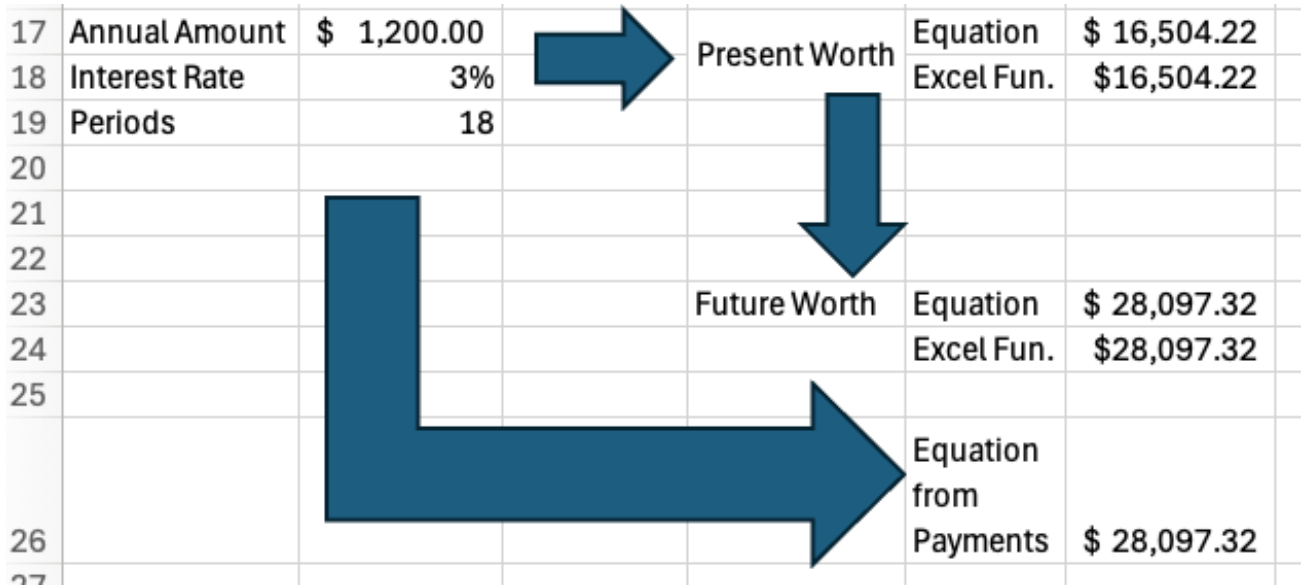
Very Important

Present Worth is timed **before** 1st annuity
Future Worth coincide with last payment

$$(F/P, i, n) = (1+i)^n$$

$$(P/A, i, n) = \left(\frac{(1+i)^n - 1}{i(1+i)^n}\right)$$

Uniform Series Factors - Excel



17	Annual Amount	$ 1,200.00	Present Worth	Equation	$ 16,504.22
18	Interest Rate	3%		Excel Fun.	$16,504.22
19	Periods	18			
20					
21					
22					
23			Future Worth	Equation	$ 28,097.32
24				Excel Fun.	$28,097.32
25					
26				Equation from Payments	$ 28,097.32

PV(rate, nper, pmt, [fv], [type])

FV(rate, nper, pmt, [pv], [type])

Uniform Series Factors – Capital Recovery (A/P) and Sinking Fund (A/F)

- Simply the inverse of P/A and F/A

$$A/P = \frac{i(1+i)^n}{(1+i)^n - 1}$$

$$PMT(rate, periods, PV)$$

$$A/F = \frac{i}{(1+i)^n - 1}$$

$$PMT(rate, periods, , \mathrm{F}V)$$



Uniform Series Factors – Capital Recovery (A/P) and Sinking Fund (A/F)

- Simply the inverse of P/A and F/A

$$A/P = \frac{i(1+i)^n}{(1+i)^n - 1}$$

$PMT(rate, periods, PV)$

	A	B	C	D	E	F	G
1	Alternative	1	2				
2	Payment	$ (200.00)	$ (220.00)				
3	i(%)	5%	4%				
4	n	20	20			**Capital Recovery**	
5						Alternative 1	Alternative 2
6	**Present Worth**						
7	Formula	$ 2,492.44	$ 2,989.87		Formula	$ 200.00	$ 220.00
8	Excel	$2,492.44	$2,989.87		Excel	$200.00	$220.00
9		*From Future Worth*					
10	Formula	$2,492.44	$2,989.87				
11	Excel	$2,492.44	$2,989.87			**Sinking Fund**	
12						Alternative 1	Alternative 2
13	**Future Worth**						
14	Formula	$ 6,613.19	$ 6,551.18		Formula	$ 200.00	$ 220.00
15	Excel	$6,613.19	$6,551.18		Excel	$200.00	$220.00
16		*From Present Worth*					
17	Formula	$ 6,613.19	$ 6,551.18				
18	Excel	$6,613.19	$6,551.18				

$$A/F = \frac{i}{(1+i)^n - 1}$$

$PMT(rate, periods, , \mathrm{FV})$

How to read and compare?

Alternative	A	B
Total out of pocket	4000	4400
Present Worth	2,494 (1506)	2,989 (1411)
Future Worth	6,613 (2613)	6,551 (2151)

Table with PW, FW and the total out of pocket. Interest is between parentheses

Example - Car Loan

- Bart is getting a Nissan GTR R34 for 200,000. He has a downpayment of 80,000 and would require a load of 120,000 to pay for his dream car. The bank offers him the amount at an **annual rate** of 7% **compounded monthly.**

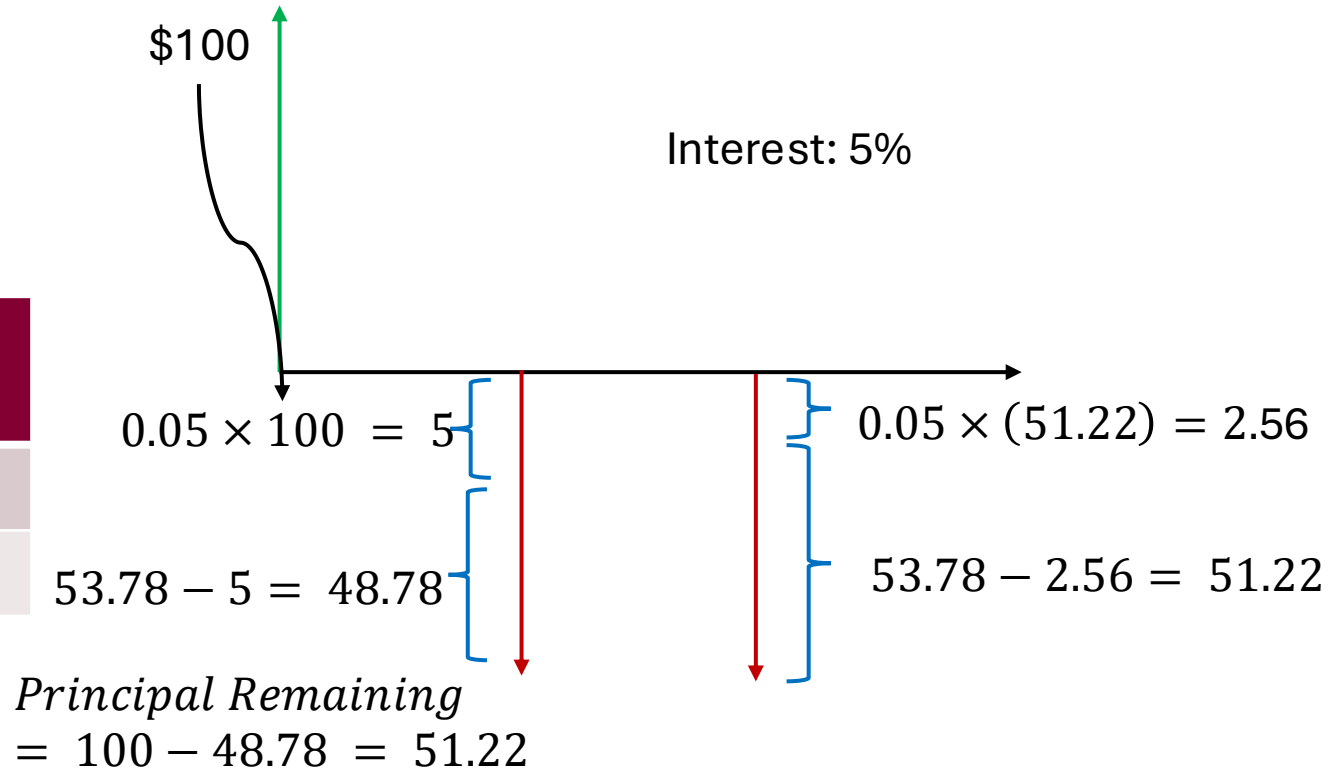
- What is the monthly payment?
- Draw the Cash flow diagram?
- How does the Bank manage the loan?



Running Amortization

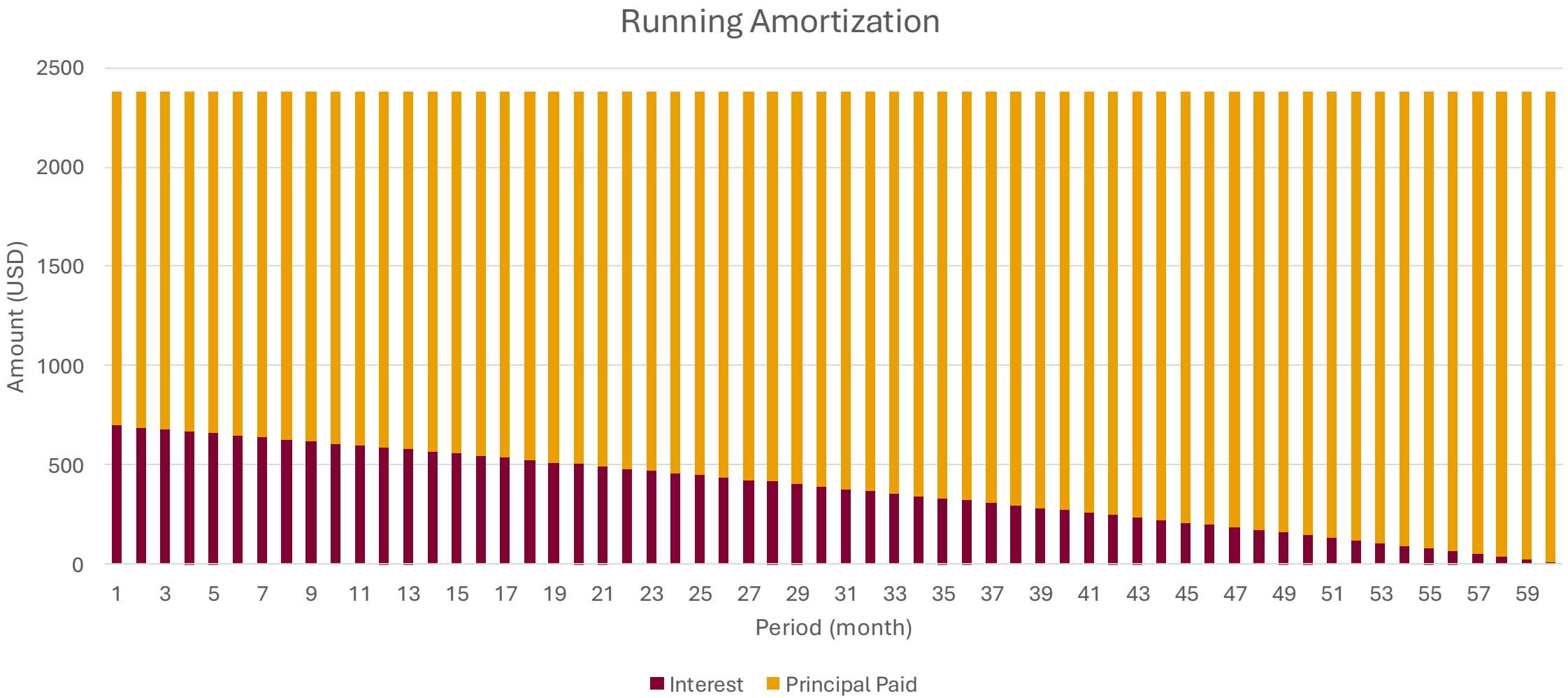
- The bank uses the payment to:
  - Partially repay the principle
  - Cover the interest

Period	Previous Balance	Payment (A)	Interest	Principal Paid	New Balance
1	100	53.78	5	48.78	51.22
2	51.22	53.78	2.56	51.22	0



A = $53.78 per period

Car Loan of Bart



Factors

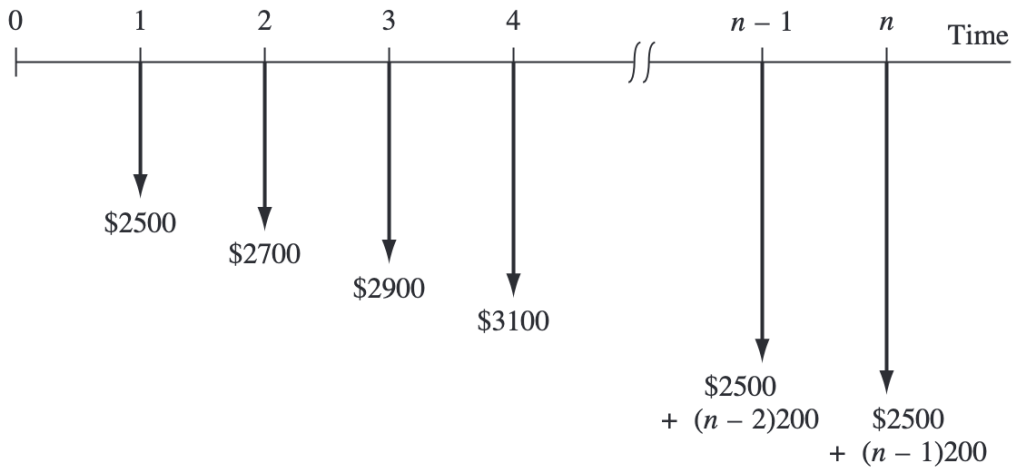
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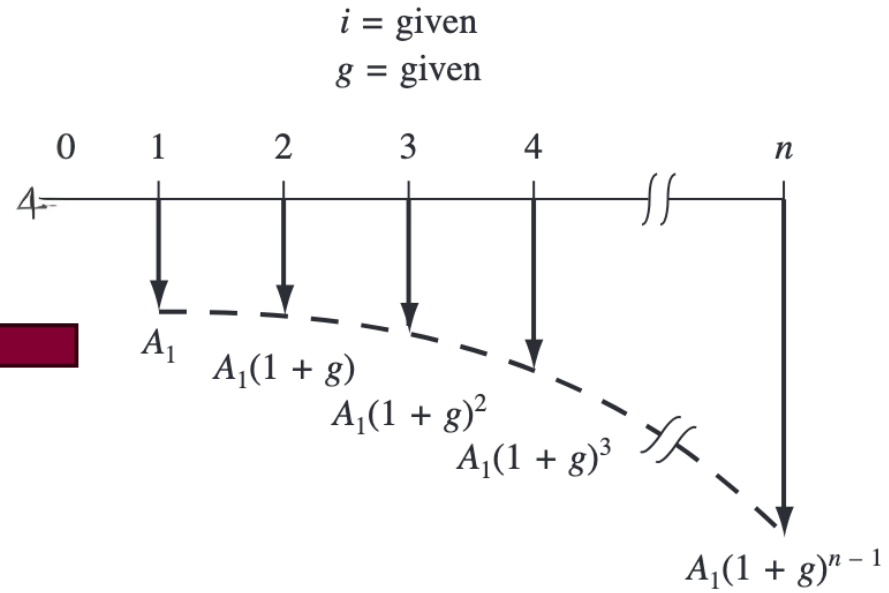


Gradient Series and Factors



Some series increase arithmetically

$$A_n = BaseAmount + (n-1)G$$



Some series increase geometrically

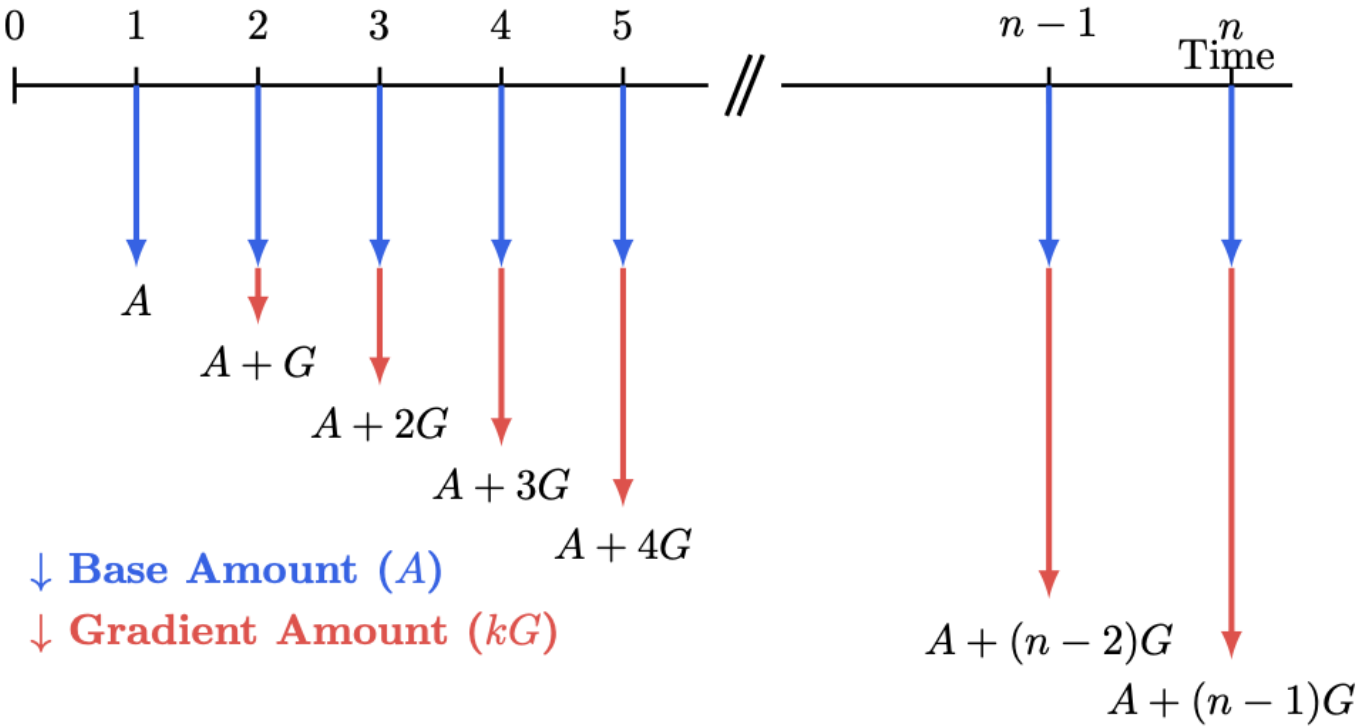
$$A_n = A_1(1+g)^{n-1}$$

Arithmetic Gradient Series

- Series that increase by a constant amount G
- It may also have a base amount: A
- Present Worth:

$$P = P_A + P_G$$

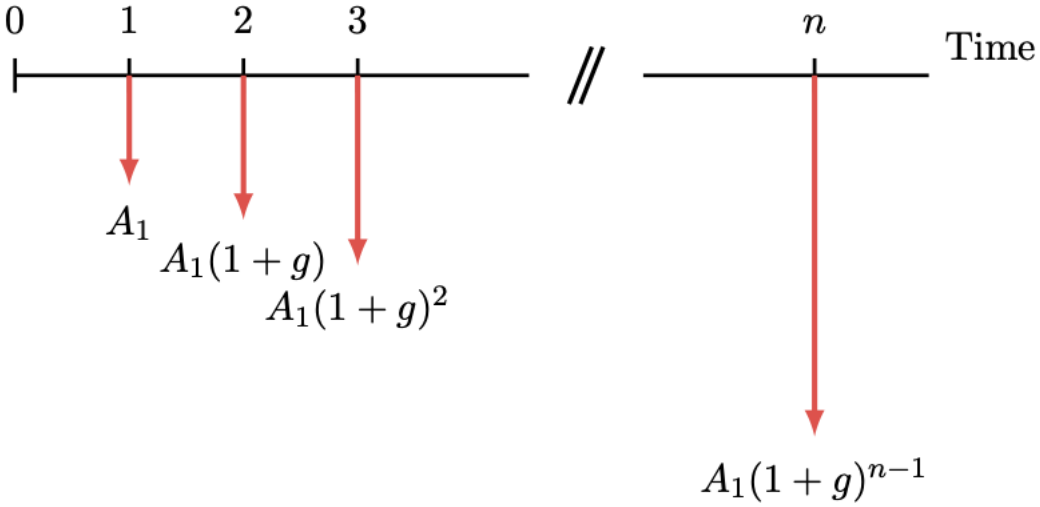
$$P_G = G\underbrace{\left(\frac{(1+i)^n - ni - 1}{i^2(1+i)^n}\right)}_{(P/G,I,n)}$$



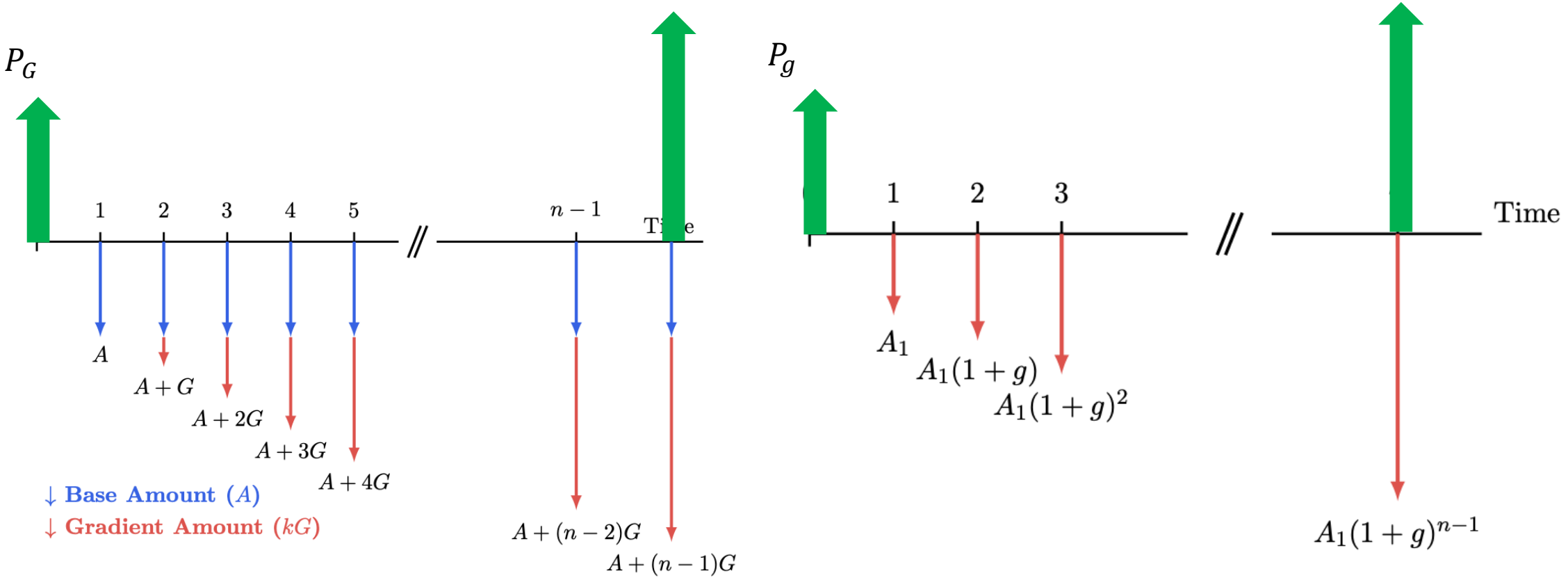
Geometric Gradient Series

- Series that increase by a constant **rate of change**: g
- It starts from an initial amount: A_1
- Present Worth:

$$(P/A, g, i, n) = \begin{cases} \dfrac{\dfrac{1+g}{1+i} - 1}{g - i} \\ \dfrac{n}{1+i} \end{cases}$$



Where to put the equivalent cash flows?



Example – Composite Cash Flow

- A company is upgrading its manufacturing facility. The upgrade has two cost components over a 10-year period:
  1. Maintenance costs for a new machine start at $1,000 at the end of Year 1 and increase by a geometric rate of 5% each year through Year 10.
  2. An additional software subscription cost of $500 per year. The first 4 years are free of charge.
- If the interest rate is 8% per year, what is the total equivalent present worth (P) of these costs at Year 0? (Begin by drawing a cashflow diagram)

